
DEVELOPMENT DOCUMENTATION

1. System Overview

What the system is, who uses it, and what it produces

Dynamic Auctioneers Marketing Platform

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Compiled 20 August 2026 · code state **9c9e0df** on main

1.1 The problem it was built to solve

Dynamic Auctioneers' property marketing was a manual bottleneck. Every ad was a hand-made artifact — a Canva poster, a portal listing, a Facebook post, an email — and every price, photo or date change meant remaking and reposting all of them, **four or more times per property**. The marketer who carried that work was leaving and was not being backfilled.

The bottleneck is structural, not a staffing problem: the data and the presentation were welded together. Every artifact held its own private copy of the price, the address and the photos, so a single change had to be applied by hand in every place it had been duplicated.

1.2 The fix

Separate the data from the presentation. One structured property record per DP number; every marketing artifact is *rendered* from that record. Change the record once, regenerate everything, push the updates. A repost becomes a two-minute approval instead of an afternoon.

A price change today reaches all nine artifacts in one pass (D64).

1.3 What was actually delivered

A **web platform** the marketing team logs into and drives themselves, at <https://46.202.175.127.nip.io> (a temporary hostname; the intended address is marketing.dynamicauctioneers.co.za). It runs the team's real workflow end to end:

drop the documents → verified record → generated advert → internal approval → client approval → full artifact set → posted to the channels → change requests and re-posts

The engine that does the work sits behind it as modules M1–M7; the platform (M8) is the face the team uses. The developer operates nothing day to day.

Secondary wins, all of them realised:

- The generated advert doubles as a mandate-pitch tool — a liquidator can be shown the campaign before signing.
- The verification step catches real data errors. Proven twice: Lightstone claimed three garages on DP3060 where the inspection found none; Lightstone said 106 m² and two bedrooms on another property where the valuer measured 310 m² and four.
- Every artifact version, approval and outbound push is logged per property, which produces "Proof of Marketing" as a by-product.

1.4 Who uses it

PERSON	ROLE IN THE SYSTEM	WHAT THEY DO
Nikki	marketing	Uploads the documents, adds photos, edits copy, drives every property through the platform
Approvers (behind <code>admin@dynamiccautioneers.co.za</code>)	approver	Sign gate 1 (verification) and gate 2 (advert). Can act straight from the approval email without logging in
Gerrie Venter	source	Prepares the Property Reports the system consumes (<code>properties.admin@</code>)
Ronnie	owner	Sets policy and rules on the open decisions
Brad	technical	Technical contact on the Dynamic side
Keegan (Cognexa)	builder	Builds and deploys. Not in the daily loop by design

Roles in the code are `admin`, `marketing` and `approver` (`webapp/auth.py`). The split exists so operational staff can run everything without ever touching credentials (D34) — the Settings screens that hold API tokens are admin-only.

1.5 What the system produces

Nine artifacts per property, all rendered from the one record (`engine/render/base.py`):

FORMAT	OUTPUT	WHAT IT IS
<code>demo_ad</code>	HTML + PNG + PDF	The branded advert. Rendered at exactly the Instagram post canvas, 1080×1350, rasterised 2× to 2160×2700 (D48)
<code>info_pack</code>	PDF	The buyer-facing information pack: a real paginated A4 landscape document rebuilt to the team's own packs (D58, D61, D67)
<code>auction_board</code>	HTML → PDF	The on-site board, rebuilt to the team's own board (D74)
<code>portal_listing</code>	Markdown	Property24-ready copy
<code>facebook_post</code>	Markdown	Facebook post plus boost notes
<code>email_blast</code>	Markdown	Subject line A/B plus body
<code>alert_mailer</code>	HTML	The alert mailer, plus the audience
<code>saia_banner</code>	HTML	SAIA alert banner
<code>webapp_icon</code>	SVG	Upcoming-auction tile for the website

Four advert designs are selectable per property — `hero_overlay` (the default), `stats_first`, `feature_list` and `collage` — which between them cover every distinct layout in the team's own 55 adverts (D41–D45, D49).

1.6 The four principles the build never bends

- 1. The DP number is the primary key.** Records, files, artifacts and leads all hang off it. Sub-properties (`DP3035.1`) are children of the instruction (`DP3035`). The DP number is *internal* — it never appears on a public artifact, where the public reference is the mandate MASTER REF (D37). The one exception is auction adverts, which restored `PROPERTY REF: DP` on the chrome (D42).
- 2. Privacy by architecture, not by discipline (POPIA).** Owner name and ID number, occupant contact, bond and arrears live only in the record's internal layer. Public renderers receive a `public_view` projection that *does not contain those fields* — it is not a matter of remembering to omit them. The same projection also strips sale-strategy figures: municipal, professional and forced-sale valuations never reach a renderer or the copy model (D54, D57). A poison-marker test asserts this on every backend and every screen. Enquiries route to Dynamic on 086 155 2288 / `properties@dynamicauctioneers.co.za`, never to the occupant.
- 3. Three human gates, enforced in code.** Nothing publishes without verification sign-off (gate 1), internal advert approval (gate 2) and logged client approval (gate 3). These are states in a machine that rejects illegal transitions, not a convention people are asked to follow.
- 4. No hallucinated facts.** Every claim in an advert traces to a record field; every record field traces to a source document or a cited piece of research. Missing data is `null`, never invented — the schema forbids fields it does not know about.

Two further rules bind the output: client-facing copy is South African English with no em or en dashes and no AI-sounding constructions, and the brand is fixed (gold `#B08D4A`, ink `#191613`, Montserrat, the real letterhead, and the Dynamic Solutions 1068 (Pty) Ltd footer with its registration and VAT numbers).

1.7 Status as at 20 August 2026

Deployment	Live on a Hostinger VPS behind nginx and Let's Encrypt TLS. Responding HTTP 200
Test suite	370 passed, 15 skipped in 52s — run and confirmed on 20 August 2026
Pipeline	Every stage has run live end to end: extraction, verification, rendering, distribution
Properties run live	The DP3060 golden case, plus a never-seen second property (Erf 2035, Somerset Park, Umhlanga) extracted at 61 of 63 fields correct on a field-by-field audit, with zero hallucinations
Build phases	0–5 and 7 complete; phase 6 (the OneDrive watcher) deliberately optional
Decision log	77 decisions logged
Outstanding	The client fix list (38 items, partially shipped), 12 decisions awaiting the owner, backups not yet configured, and the real domain not yet cut over

Detail on all of it is in section 6.

1.8 What it costs to run

Roughly **R20 per property** in AI spend — extraction about R5, verification with web research about R10, copy generation about R3. Re-renders on updates cost cents, because the two paid calls are content-addressed cached (D59) and the stable prompt prefix is cached (D32). That is noise against the roughly R6,000 per month tool budget the system is meant to consolidate into. The VPS is the only fixed cost.

1.9 Glossary

TERM	MEANING
DP number	Dynamic Auctioneers' property file number (<code>DP3060</code>); sub-properties <code>DP3060.1</code>
EVM	Lightstone's Estimated Value Model report — desktop valuation, deeds data, comparables
Property Report	Dynamic's in-house branded inspection document, prepared by Gerrie
OTP	Offer to Purchase / Conditions of Sale — the document the sale terms are read from
Lightstone	South African property data provider
Prop Data	South African property-portal syndication backbone, feeds Property24
GHL	GoHighLevel — the Social Planner used for posting to Facebook, Instagram and LinkedIn
POPIA	Protection of Personal Information Act, South African privacy law
Instruction	The engagement or liquidation under which one or more properties are sold
Forced-sale value	A valuer's quick-disposal estimate, typically 20–40% below market. Internal only
SAIA	South African Institute of Auctioneers — industry body, and the alert-mailer audience
Gate	A point where a human must approve before the system proceeds
<code>public_view</code>	The projection of a record with private and strategy fields physically removed